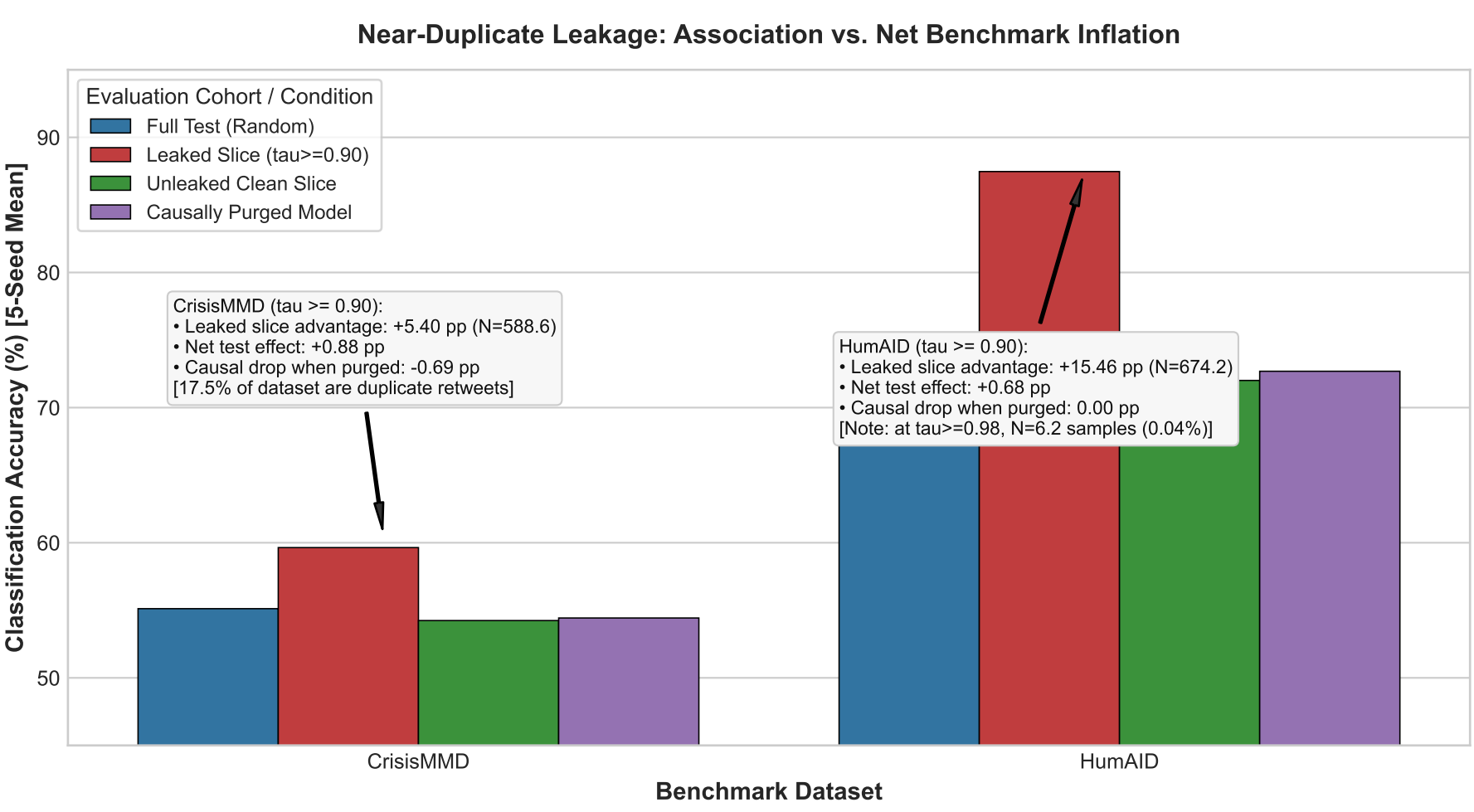
Near-Duplicate Leakage: Association vs. Net Benchmark Inflation
Evaluation Cohort / Condition
Full Test (Random)
Leaked Slice (tau>=0.90)
Unleaked Clean Slice
Causally Purged Model
CrisisMMD (tau >= 0.90):
• Leaked slice advantage: +5.40 pp (N=588.6)
• Net test effect: +0.88 pp
• Causal drop when purged: -0.69 pp
[17.5% of dataset are duplicate retweets]
HumAID (tau >= 0.90):
• Leaked slice advantage: +15.46 pp (N=674.2)
• Net test effect: +0.68 pp
• Causal drop when purged: 0.00 pp
[Note: at tau>=0.98, N=6.2 samples (0.04%)]
Classification Accuracy (%) [5-Seed Mean]
90
80
70
60
50
CrisisMMD
HumAID
Benchmark Dataset